

Pricing NHL Trades on One Scale

RESEARCH QUESTION

When NHL teams trade players, draft picks, and prospects, do they price each kind of asset consistently, or do some kinds reliably sell for more or less than they turn out to be worth?

1. The Idea

A hockey trade is a swap of very different things: a player on a signed contract, a draft pick that might become anyone, or a teenager playing junior hockey. Comparing them is like comparing a car, a lottery ticket, and an apprentice, so the project converts each one into the same currency, which is the *surplus value* it gives its team in dollars. Surplus is what the player's performance over time would cost to buy on the open market, minus what the team actually pays him under the salary cap. Once each asset is in the same currency, the two sides of any trade can be added up and compared, and then checked against what actually happened on the ice.

What his play is worth the going market price for the wins he is expected to add	—	What he costs his salary cap hit	=	Surplus value the bargain (or overpay) the team is holding
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For example, a player whose play would cost \$6 million a season on the open market but who is paid \$4 million gives his team \$2 million of surplus that season. The model adds this up over each remaining season, allowing for the chance he declines or leaves the league, and counts a dollar of surplus in a later season for less than a dollar today.

2. The Three Kinds of Assets

ASSET 1 Rostered players <i>Like buying a used car with a known price tag.</i> An NHL player on a signed contract. His cost is known (his cap hit); what is uncertain is how well he will play. If he cannot yet sign anywhere when the contract ends (restricted free agent), the team keeps his rights for a few more years.	ASSET 2 Draft picks <i>Like a lottery ticket; higher picks have better odds.</i> The right to choose a player at a given spot in a future draft. Most late picks never play an NHL game, but the few that do can be very valuable, because drafted players start on cheap entry-level contracts and teams have control over their contract rights.	ASSET 3 Prospects <i>Like an apprentice who may or may not make the grade.</i> A player already drafted but not yet in the NHL, playing in junior, college, the minors, or Europe. Each season he plays tells us more about whether he will make it and how good he will be.
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3. What the Project Looks For in the Data

1. **Value each trade on the day it was made.** For each trade from 2018 to 2026, the model prices what each team gave up and what it received, using only information available on that date.
2. **Compare the two sides.** Each trade is expressed as a ratio of value sent to value received. A fair trade sits near 1.
3. **Follow what actually happened.** The prediction is then compared with how the players performed afterward.
4. **Look for patterns.** The question is whether mistakes are random or systematic, for example, whether picks are routinely overpaid for, or whether long contracts, older players, or contending teams buying at the deadline consistently come out behind.

Keeping the test honest. Three rules protect the results:

- **No peeking.** A valuation uses only what was known before the trade date.
- **A separate ruler.** Outcomes are also checked with a measure built independently from NHL game data.
- **Count the misses.** Averages include each pick and prospect who never made it, not just the successes, so the odds are not overstated.

If no pattern appears, that is also a finding: it would suggest teams price these assets sensibly on average.

4. How the Model Values Each Asset

Each kind of asset follows its own path to a dollar figure. All three paths end in the same currency, so they can be combined into a single value for each side of a trade.

